

Question:

Eliza's rate per hour for the first 40 hours she works each week is \$10. She also receives an overtime pay of 1.2 times her regular hourly rate. If Eliza worked for 45 hours this week, how much are her earnings for this week?

Reasoning + Answer:

Let's break down the problem step by step .

1. **Calculate Eliza's regular pay for the first 40 hours : **

- Eliza's regular hourly rate is \$ 10 .

- For the first 40 hours , her pay is calculated as :

$$\begin{aligned} & 40 \text{ \text{ hours } } \times \$ 10 \text{ \text{ hour } } = \$ 400 \end{aligned}$$

2. **Calculate the number of overtime hours : **

- Eliza worked a total of 45 hours this week .

- The number of overtime hours is :

$$\begin{aligned} & 45 \text{ \text{ hours } } - 40 \text{ \text{ hours } } = 5 \text{ \text{ hours } } \end{aligned}$$

3. **Calculate Eliza's overtime pay rate : **

- The overtime pay rate is 1 . 2 times her regular hourly rate .

- Therefore , the overtime pay rate is :

$$\begin{aligned} & 1 . 2 \times \$ 10 = \$ 12 \text{ \text{ hour } } \end{aligned}$$

4. **Calculate Eliza's pay for the 5 overtime hours : **

- For the 5 overtime hours , her pay is :

$$\begin{aligned} & 5 \text{ \text{ hours } } \times \$ 12 \text{ \text{ hour } } = \$ 60 \end{aligned}$$

5. **Calculate Eliza's total earnings for the week : **

- Adding the regular pay and the overtime pay , we get :

$$\begin{aligned} & \$ 400 + \$ 60 = \$ 460 \end{aligned}$$

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- Eliza earns \$ 10 per hour for the first 40 hours , which totals to \$ 400 .

- For the 5 overtime hours , she earns 1 . 2 times her regular rate , which is \$ 12 per hour , totaling \$ 60 .

- Therefore , her total earnings for the week are \$ 460 .

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<answer>\$ 460 </answer>